

Authors Contributions

MF designed and implemented the analysis pipeline, performed all modeling and statistical analysis, produced all figures and tables, including those in Multimedia Appendix 1, and wrote the Methods and Results. MP directed the project, supervised its scientific design and interpretation, and wrote the rest of the manuscript. KLF prepared the upstream clinical data, including the treatment-resistant depression labels and the antidepressant index dates that define the cohort and the outcome. RK, SS, and DP provided methodological feedback and guidance on the analysis and its interpretation throughout. All authors reviewed and approved the final manuscript.